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A Palaeomagnetic Study of the Gaberones Granite of Botswana

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Summary

Thirty-two cores were drilled at eleven sites in a variety of granitic rock-types. Meaningful results were obtained from seven sites after alternating field demagnetization and there is good evidence for the stability of the magnetization observed. The specimens exhibit a wide range in initial intensity of magnetization but no relationship exists between rock-type, initial intensity and behaviour under a.f. demagnetization. In all cases the within-site scatters are large and the possibility that this is a fundamental property of the remanent magnetization of large granitic bathyliths is discussed.

The location of the mean palaeomagnetic pole for the Gaberones granite (103° E, 34° N, $A_{95} = 16°$) is compatible with the smooth polar wandering path deduced from other ancient African poles. It is also consistent with the radiosotopic age for the granite of 2350 million years.

1. Geology

The Gaberones granite occurs in the southeastern corner of the Republic of Botswana (previously the Bechuanaland Protectorate) on the border of South Africa (Fig. 1). It is exposed as a number of large plutons (Poldervaart 1952) which consist of a central rapakivitic granite surrounded by two border phases comprising an inner zone of red or grey granite (Marginal granite) and an outer zone of porphyritic granophyres (Wright 1958). The Central granite is usually porphyritic and contains large phenocrysts of white felspar with idiomorphic or ovoidal outlines and in large outcrops it is seen to be foliated. The Marginal granite is typically a coarse-grained equigranular rock almost entirely composed of anhedral red or grey felspar and quartz. Beyond the Marginal granite, the porphyritic granophyres show a perfect transition into fine-grained felsites which were previously correlated on lithological grounds with the Dominion Reef System of South Africa.

In recent years a number of radioisotope age determinations have become available and some of these ages have led the Botswana Geological Survey to tentatively regard the felsites as pre-Dominion Reef System in age. In view of this uncertainty. the maximum age of the Gaberones granite is set by the age of the Modipe gabbro into which it is intrusive. This gabbro has been dated by McElhinny (1966a) who obtained an age of 2630 ± 470 million years from Rb : Sr decay.

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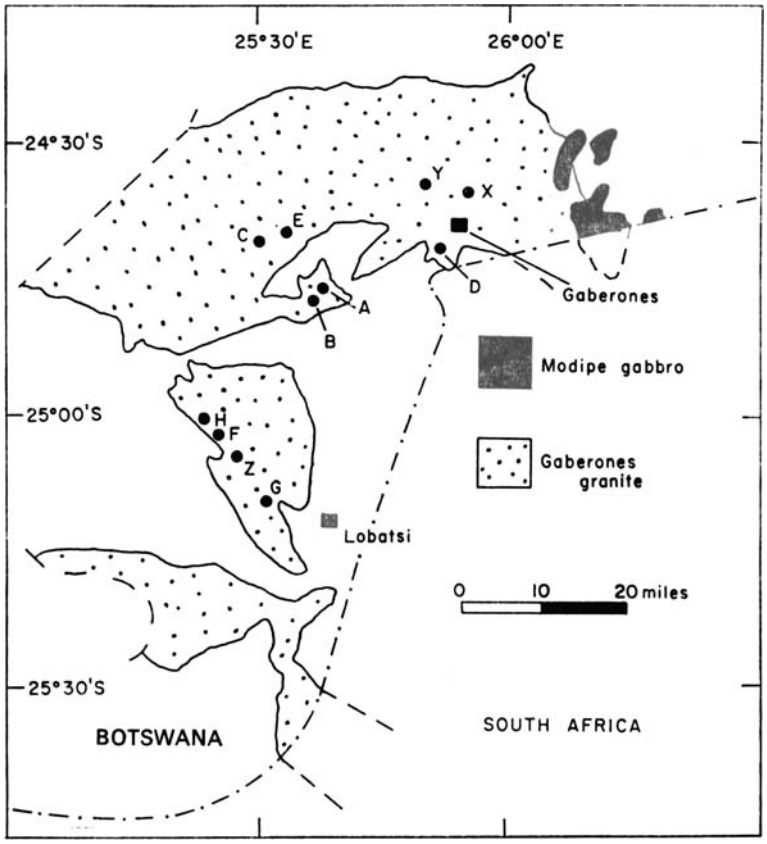


FIG. 1. Geological sketch map of southeastern Botswana showing the eleven sampling sites.

Jones (1963) regards the granite as pre-Transvaal System and possibly pre-Ventersdorp System in age, although the Geological Survey indicate it as pre-Ventersdorp System on the 1966 provisional map of Botswana. Table 1 lists a number of radioisotope ages for the granite. A decay constant of $1{\cdot}39 \times 10^{-11}$ year^{-1} was used in all Rb : Sr determinations but only that of McElhinny (1966a) was calculated using the isochron method and is therefore thought to be more reliable.

2. Sampling and measurement

Samples were collected by drilling since exfoliation of the granite usually leads to well rounded ' whaleback ' outcrops from which it is impossible to remove reasonably sized samples by hand. A total of thirty-two cores, each about 2 ft long, was drilled from eleven sites (Fig. 1), using a modified version of the equipment described by McElhinny & Neal (1966). All the cores were oriented using the method of Gough & Opdyke (1963). From each core, a number of cylindricai specimens (1 inch long, 1 inch diameter) were cut and the natural remanent magnetizations (NRMs) were measured on a spinner magnetometer (Gough 1964).

Results from two of these sites have been reported by Evans & McElhinny (1966) in a preliminary study of the granite. Upon initial measurement, the NRM directions at seven sites were random according to the criterion of Watson (1956). All specimens were therefore subjected to alternating field demagnetization in the

Table 1

Radioisotopic age determinations of the Gaberones granite

Method	Age in million years	Source
Rb : Sr whole rock	2980±100	Bechuanaland Geological Survey
Rb : Sr whole rock	2880±100	Annual Report for 1965
K : Ar on hornblende	2305±115	Analyst: N. J. Snelling
Rb : Sr whole rock isochron	2340±50	McElhinny (1966a)

apparatus described by McElhinny (1966b). Specimens were very variable in initial intensity of magnetization ranging from 19·0 $A.m^{-1}$ ($1{\cdot}90 \times 10^{-2}$ e.m.u. cm^{-3}) to below the lower limit of the magnetometer ($1{\cdot}50 \times 10^{-7}$ e.m.u. cm^{-3}). The a.f. demagnetization characteristics were also very variable, some specimens showing ' end-points ' and some merely grouping at a particular stage and then diverging. There is no relationship between rock-type, initial intensity and behaviour under demagnetization; useful results were obtained from both weak and strong specimens of all rock-types (Table 2).

After demagnetization seven of the sites gave meaningful results but the directions of magnetization at the remaining four sites were still randomly scattered. Of these ' random ' sites one in particular (site D) is thought to have been struck by lightning since the specimens are all strongly magnetized (2–4 $A.m^{-1}$) and the individual specimens vectors, although widely scattered, were very stable under a.f. demagnetization. Sites B and C are situated in the Lemau and Sephatlaphatla Hills respectively and the poor results obtained from these sites may be due to the strong foliation which is reported by Wright (1958) in these areas.

The results obtained after magnetic cleaning from the seven sites are given in Table 2. One anomalous specimen was rejected from each of sites G and X since their directions of magnetization were more than $140/k^{\frac{1}{2}}$ degrees from the corresponding site mean direction. This means that there is less than 5% probability that they actually belong to the site populations in question.

The results obtained are regarded as being reliable since the mean direction of magnetization for the seven sites (Table 2) lies 75° from the direction of the present geomagnetic field at the sampling sites. The most powerful stability test, however, is the ' baked-contact ' test reported by Evans & McElhinny (1966). In this test it was shown that three samples of Modipe gabbro collected near the contact with the Gaberones granite had NRM directions in agreement with the direction of magnetization of the granite itself thereby demonstrating both the reheating of the gabbro by the granite and the stability of the thermoremanences acquired by both rocks at that time.

Even after improved grouping under a.f. demagnetization, the precisions (k) of the measurements at each site are rather low. It is possible that these low precisions may be a fundamental property of the remanent magnetization of granitic rocks.

3. Sources of dispersion

In a study of a group of large granitic bathyliths in California, Currie *et al.* (1963) also observed low within-site precisions and found that more than half of their sites were still randomly magnetized after attempts at magnetic and thermal cleaning. These authors discuss in detail the possible sources of dispersion and they conclude that the behaviour of their specimens under a.f. demagnetization indicates the presence of a ' random ' component of magnetization of unknown origin.

If the granite cooled very slowly then individual specimens at a site may be sufficiently different in ' magnetic ' age for secular variation to have contributed to

Table 2

Site mean directions of magnetization for the Gaberones granite

Site	Rock-type	Site position		C	N	B_{ac}	D	I	R	k	Site pole		
		Lat. S	Long. E								Lat. N	Long. E	N or S
A	Central granite	24° 45′	25° 39′	3	6	80	33½	−53	4·963	4·8	59·5	140·1	N
E	Central granite	24° 41′	25° 33′	3	5	60–100	55½	+27	4·120	4·5	23·1	86·1	N
F	Central granite	25° 03′	25° 25′	3	6	60–120	64	+26½	5·018	5·1	16·7	90·8	S
G	Marginal granite	25° 11′	25° 33′	3	5	5–10	34½	−24½	4·684	12·6	55·1	101·4	N
X	Microgranite	24° 36′	25° 56′	3	7	10	60½	−11½	6·034	6·2	29·2	108·7	N
Y	Marginal granite	24° 36′	25° 51′	2	9	10–40	61	+ 3½	8·249	10·7	25·3	101·3	N
Z	Central granite	25° 06′	25° 28′	2	5	10–20	64½	−17	4·690	12·9	26·9	112·6	N
Mean of seven site directions							57½	−7	6·148	7·04	33·3	102·9	

N, number of specimens
B_{ac}, peak demagnetizing field in $mWb.m^{-2}$
R, resultant of N unit vectors
C, number of cores

D, declination in degrees east of north
I, inclination in degrees (positive below horizontal)
k, best estimate of precision (see Fisher 1953)

the within-site dispersion. Jaeger (1957) has given curves which describe the cooling history of a granitic sheet of thickness D metres intruded at 700 °C into country rock at 0 °C. These curves show that the centre lags behind a point situated $0{\cdot}1D$ m from the edge by $0{\cdot}008D^2$ years in cooling to 600 °C. The average rate of displacement of the 600 °C isotherm is thus $0{\cdot}4D/0{\cdot}008D^2 = 50D^{-1}$ m/year, which gives 0·5 cm/year for $D = 10$ km and 0·05 cm/year for $D = 100$ km. Despite the fact that the Gaberones granite is not a sheet-like intrusion, these values serve for an order of magnitude calculation. Since the cores at a site are often 5 m apart their ' magnetic ' ages may differ by 10^3 to 10^4 years, during which time a considerable amount of secular variation may take place (Cox & Doell 1960).

Each specimen will probably have a range of blocking temperatures, say from 600 °C to 500 °C. The time taken by a specimen to cool through this range is of the order of 10^5 years (Jaeger 1957, Currie *et al.* 1963). Thus unless the blocking temperature spectra are sharply peaked the ' magnetic ' age of each specimen will be a range of ages which overlaps 90% or more with the age ranges of all other specimens from the same site. It is unlikely, therefore, that secular variation is an important source of within-site dispersion, although it will almost certainly contribute to the between-site dispersion.

Granitic magma containing volatiles may remain molten below 600 °C (Tuttle & Bowen 1958), and it is therefore possible for movement to occur within the range of blocking temperatures. This could cause rotation of some of the magnetic vectors away from the ambient field and may be the origin of the ' random ' component described by Currie *et al.* (1963). The vectors involved in this rotation would belong to the higher temperature partial TRMs, and, although possibly random in direction, would be very stable (Néel 1949). In extreme cases such rotation might result in the *total* NRMs at a site becoming randomly oriented.

4. Palaeomagnetic pole position

Fig. 2 is a stereographic projection of the seven site mean directions of magnetization. Six of the sites have normal polarity (the same sense as the present field), but the magnetization at site F is approximately reversed. The mean vector for site F lies 138° from the mean of the other six sites, and its reverse thus lies 42° from this

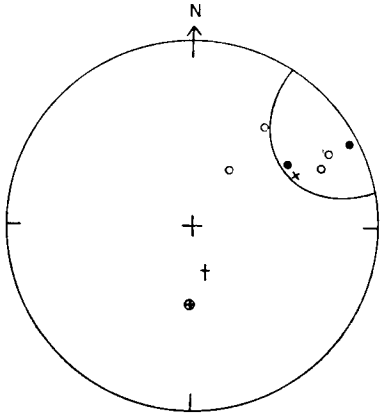


FIG. 2. Stereographic projection of seven site mean directions of magnetization and the circle of 95% confidence for the mean (upper hemisphere part only). ●, north-seeking direction down; ○, north-seeking direction up; ×, south-seeking direction down; †, present field, south seeking; ⊕, axial dipole field, south seeking.

Table 3

Statistical parameters for the two-tier analysis

D	I	N	B	ω	β	k^*	α_{95}
55½	−6½	43	7	6·87	11·21	62·0	18

B, number of sites
ω, within-site precision parameter
β, between-site precision parameter
$k^*=[(\omega N)^{-1}+(\beta B)^{-1}]^{-1}$ precision parameter for mean of forty-three unit vectors
$\alpha_{95}=140/k^*$, semiangle of cone of 95% confidence for mean direction
Other symbols as in Table 2.

mean. The probability that a direction will be observed which makes an angle of 42° or more with the true mean direction is 0·14 (Watson & Irving 1957), which means there is no reason to suppose that site F is not reversely magnetized with respect to the other six sites. Thus in calculating the mean direction of magnetization of the seven sites, the mean direction of site F has been rotated through 180°. In any case the inclusion of site F changes the mean direction by 6° only. The parameters of the mean direction are given at the bottom of Table 2.

The results have also been analysed by the two-tier method of Watson & Irving (1957) since it is inevitable that the between-site precision is masked by the low within-site precisions. Although this method is valid for small dispersions only, it will undoubtedly yield a better estimate of between-site precision than would be obtained by simply analysing the site mean directions. The precision, β, so obtained provides an estimate of the angular dispersion $\Delta=81/\beta^{\frac{1}{2}}$, which is a measure of the palaeosecular variation. The results of the two-tier analysis appear in Table 3 and they give $\Delta=24{\cdot}2°$. The use of an F-test (Gough *et al.* 1964) gives 95% confidence limits such that $14{\cdot}8°\leqslant\Delta\leqslant 39{\cdot}7°$. Thus the angular dispersion of the geomagnetic field at a palaeolatitude of 3·5° in Gaberones granite times was not significantly different from that calculated by Creer (1962) for the present field at the same latitude.

It will be noted from Fig. 2 that the site mean directions have a much wider range in inclination than in declination and lie approximately in a vertical plane whose

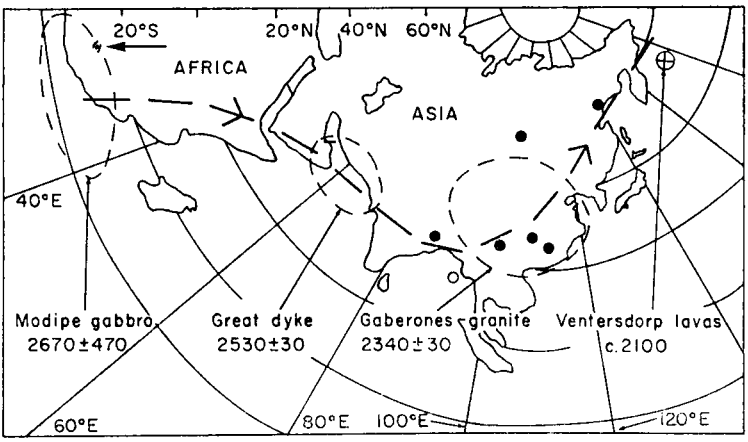


FIG. 3. Suggested polar wander curve for the ancient Pre-Cambrian of southern Africa. The seven site poles for the Gaberones granite are shown and the small arrow indicates the granite outcrop. Polar azimuthal equidistant projection. ●, north pole; ○, south pole.

bearing is 55° east of north (the declination of the mean). This is consistent with the observations of Creer *et al.* (1959) who demonstrated that the mean directions of magnetization of sites in low palaeoloatitudes can be expected to have an oval distribution such that the variation in inclination is approximately twice the variation in declination.

Since the sites themselves are wide-spread geographically it is preferable to compute the mean palaeomagnetic pole from the seven site poles rather than the mean direction of magnetization. The seven site poles, calculated on the assumption of a geocentric dipole field, are listed in Table 2 and their mean lies in central China at longitude 103½° E, latitude 34½° N with $A_{95}=16°$.

Fig. 3 shows the spatial relationship between the palaeomagnetic pole of the Gaberones granite and those of the Modipe gabbro (Evans & McElhinny 1966), the Great Dyke (McElhinny & Gough 1963) and the Ventersdorp lavas (Jones *et al.* 1967). Assuming that polar wander during this part of the Pre-Cambrian followed a reasonably smooth path, then the location of the Gaberones granite pole is compatible with the suggestion that the granite is pre-Ventersdorp System in age (Jones 1963). It also indicates that the granite most probably post-dates the Great Dyke of Southern Rhodesia, which supports the younger of the granite ages given in Table 1.

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